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Palaeomagnetism of the Jurassic Navajo Sandstone from South-western Utah

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A combination of petrographic and magnetic studies indicates that the Navajo Sandstone carries a stable secondary magnetic moment and arguments are given for believing this to have been recently acquired. It therefore follows that haematite is more mobile in permeable sandstones than has generally been thought.

The results suggest that the Jurassic palaeomagnetic pole for North America, based largely on preliminary work by Runcorn and Collinson & Runcorn from the same formation and similar red-bed lithologies, is based on rocks with a present-day component of magnetization. The virtual poles reported by DeBoer or Helsley & Steiner may be the best representations of the North American Jurassic palaeomagnetic pole position.

A petrographic study indicates that the investigation of the Navajo Sandstone as a means of detailing the behaviour and detailing a part of the polar wandering curve for North America is not feasible. However, a residuum of the original Jurassic component may be seen in the reversely magnetized sites. There probably exist similar residue in the normally magnetized sites, but these Jurassic residues are much less clearly defined.

Introduction

The investigation of the Lower Jurassic Navajo Sandstone of western North America was begun with the object of providing reliable palaeomagnetic data for this interval of geologic time, and to detail as closely as possible the behaviour of the geomagnetic field. Recently attempts have been made to identify a Jurassic magnetic anomaly sequence in oceanographic work without any real indication from palaeomagnetism of how the field actually behaved (Vogt *et al.* 1970; Larsen & Chase 1972). Palaeomagnetic control for the Jurassic is poor although McElhinny & Burek (1971) have attempted to combine palaeomagnetic results from the literature to derive a palaeomagnetic reversal chronology for the whole of the Mesozoic. Since most palaeomagnetic work has not been based on closely-spaced stratigraphic sampling, there may be many reversals that have not been reported.

Geology

Accumulated palaeomagnetic experience has shown that igneous rocks and red-bed

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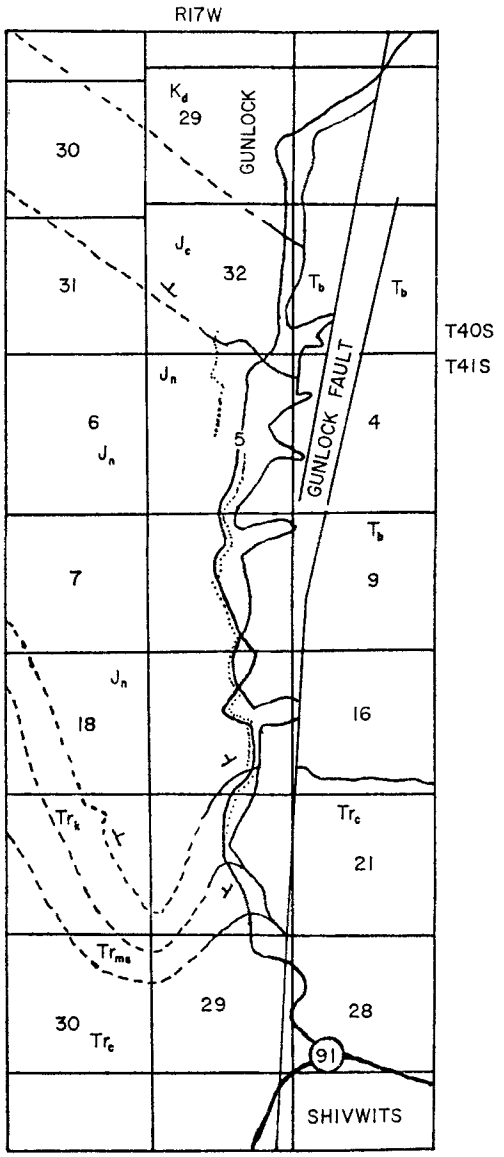


FIG. 1. Map showing approximate sampling locations for the Gunlock section. Tertiary, T_b, basalt; Cretaceous, K_d, Dakota formation; Jurassic, J_c, Carmel formation, J_n, Navajo sandstone; Triassic(?), Tr_k, Kayenta formation, Tr_{ms}, Moenave formation; Triassic, Tr_c, Chinle formation.

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sediments are suitable for palaeomagnetic work (Irving 1964). Although, in many ways, lava sequences provide one of the best means of studying the history of the geomagnetic field (DeBoer 1968), in the Jurassic of North America unaltered sequences are lacking. Since it is also impossible to use slow-cooling intrusives (Ito, Dunn & Fuller 1970) to cover the whole of the Lower Jurassic; the study of the Lower Jurassic is restricted to red-bed sequences.

Search for suitable sampling locations is constrained by the known stratigraphy of the United States (Dunbar 1962, p. 294). Lower Jurassic rocks do not crop out east of the Great Plains (Stanley, Jordan & Dott 1971), and the need to avoid rocks which have been metamorphosed leads to the exclusion of probably the most complete sections found in California. In order to have stratigraphic control normally absent in continental red-bed sequences, it is necessary to seek a region near the continental margin where intercalated marine transgressions will provide the reference horizons. The region where these conditions are met best are in southern Utah and northern Arizona (Harshbarger, Repenning & Irwin 1957; Dunbar 1962). The succession of Lower Jurassic and related rocks used in this study is (oldest to youngest): Kayenta Formation, Navajo Sandstone, and the ' Temple Cap member ' of the Carmel. Pilot collections of Upper Jurassic formations from the San Rafael Swell were made with a view to extending the survey.

The Lower Jurassic Navajo Sandstone sequence was collected near Gunlock, Utah in sections 5, 8 and 17 of T41S, R17W and section 32 T40S, R17W (Fig. 1). The stratigraphic sequence is located within the Basin and Range province (Kurie 1966), and exposures are controlled by Basin and Range structures. All sites were from the downthrown block on the west side of the Gunlock fault which has a maximum displacement of 4000 ft (Dobbin 1939). Normal drag on the west block produced the asymmetric Shebit syncline (Cook 1960). The 189 sampling sites comprising the Gunlock section were taken from both limbs of this fold. The lower 120 sites were on the east side of the fold where the strata tilt westward and north-westward at 42–14 degrees: the upper 69 sites were collected on the west limb where the strata all had north-east tilts of 8–14 degrees. The sequence is well exposed, only some 200 m in a succession of 1000 m were covered by slumped material. Tertiary basalts, mostly confined in outcrop distribution to west of the Gunlock fault, rest directly upon eroded Jurassic formations.

The Upper Triassic (?) *Kayenta* (Lewis, Irwin & Wilson 1961) at Gunlock, at the base of the section, is characterized by slope forming siltstone and claystone. The environment of deposition was that of an alluvial fan with local sediment transport directions to the south-west (Harshbarger *et al* 1957; Wilson 1965; Poole 1961). Quiet water floodplain deposits predominate at Gunlock, grading eastward to more typical Kayenta lithology (Wilson 1965). The 13 sampling sites from the Kayenta are non-uniformly distributed; the lowest three sites were separated from each other by large unexposed intervals.

The overlying *Navajo Sandstone*, characterized by massive trough and wedge-planar cross-bedded units of fine-grained sandstone, is readily differentiable from the Kayenta. The contact appears to be gradational over a one hundred foot stratigraphic interval. The gradational zone is characterized by a subtle increase in the number and thickness of cross-bedded sandstones in the upper Kayenta. Since thin beds of Kayenta-like lithology were present throughout the Navajo section, the Kayenta-Navajo contact was arbitrarily placed at the base of the first massive cross-bedded sandstone. The Navajo was sampled at an average interval of 3 m.

In contrast to other exposures of the Navajo east of St George, Utah, where the Navajo intertongues with the Carmel Formation (Thompson & Stokes 1970), the Navajo at Gunlock completely underlies the lowest known member of the Carmel. The informally designated ' Temple Cap member ' of the Carmel (Cook 1960), a facies of the Temple Cap Member of the Navajo Sandstone (Thompson & Stokes 1970),

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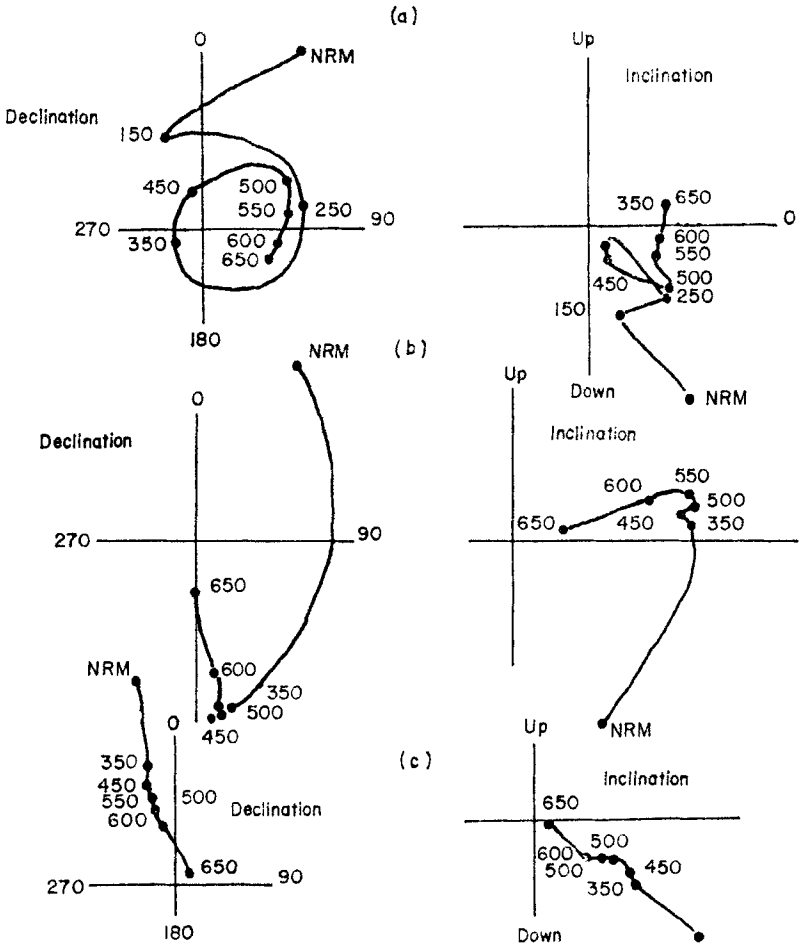


FIG. 2. (a) Unstable; (b) stable reversed direction; (c) stable normal directions. These diagrams all show direction versus normalized intensity.

underlies the Kolob Limestone Member of the Carmel whose lowest beds mark the initiation of Bajocian (Middle Jurassic) deposition in southern Utah (Imlay 1964; Sohn 1965). This means that the ' Temple Cap member ' is older than Bajocian. The contact between the ' member ' and the Kolob appears to be conformable. Since the ' Temple Cap ' is not fossiliferous, its exact age placement is therefore doubtful; however, assuming a sedimentation rate for the ' member ' similar to that calculated for the Navajo places the lower boundary of the member as middle upper Toarcian.

Palaeomagnetic technique

The field collection of oriented cores was made using a portable field drill (Cox & Doell 1967). At each sampling site four to eight cores were collected over a 40–70 cm stratigraphic interval. The lateral extent of each site ranged up to 16 m, but most of the collections were over shorter distances because of bedding tilt and limited outcrop accessibility.

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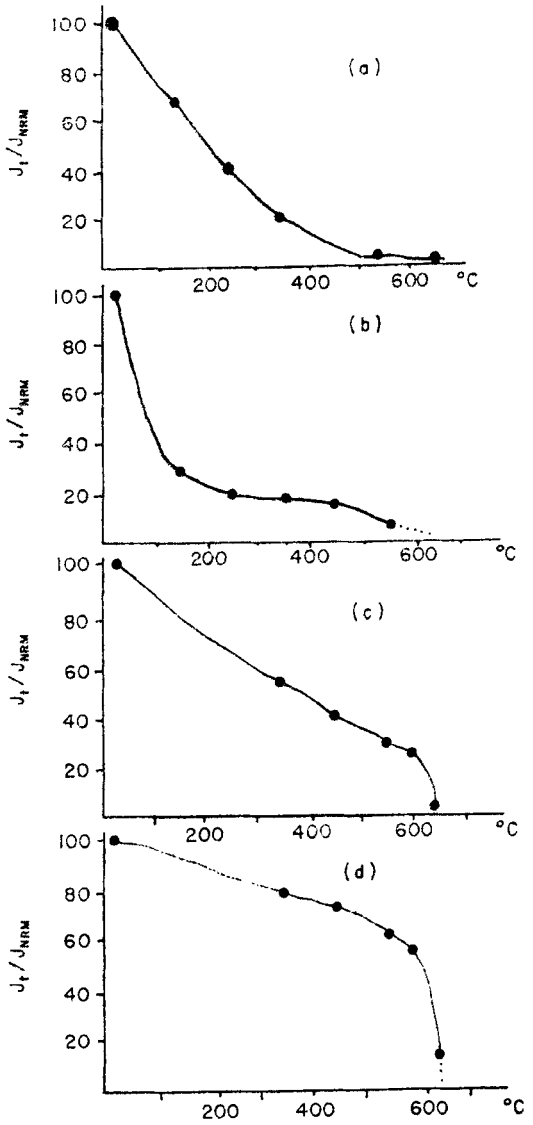


FIG. 3. Thermal demagnetization curves, all normalized. (a) Demagnetization curve for unstable (haematite stained) red beds; (b) demagnetization curve for yellow beds (limonite stained); (c) demagnetization curve for stable red beds (haematite stained—small direction changes from NRM directions; (d) demagnetization curve for ultrastable red beds (haematite stained)—no change in direction from NRM direction.

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The cores were measured on a slow type fluxgate spinner magnetometer (Foster 1966) that can reproducibly measure intensities as low as 2×10^{-7} emu cm^{-3}.

One pilot specimen was chosen from each site for step-wise thermal demagnetization at 50 °C temperature intervals, and the demagnetization data analysed for stability by a technique devised by As & Zijderveld (1958). After the pilot specimens were examined, the remaining specimens were then thermally demagnetized at three temperature intervals bracketing the optimum temperature interpreted from the preceding technique.

In order to determine when the secondary effects have been removed, Irving, Stott & Ward (1961) used the criteria of minimum dispersion of sample directions for a site at each demagnetization of sample directions for a site at each demagnetization step. More recently Briden (1972) and Symons & Stupansky (1974) have developed indices of stability based on a comparison of two vectors from different demagnetization steps. However, a quick graphical index of stability has been developed by As & Zijderveld (1958), and this index was found suitable for this study. When the As & Zijderveld criteria for stability was used, two groups become apparent. One group never reaches a stable end point (Fig. 2(a)) and commonly shows a thermal demagnetization curve similar to Fig. 3(a) and (b). Specimens which behave in this manner are considered to be unreliable and are excluded from palaeomagnetic pole determinations. The second group (Fig 2(b) and (c)) shows a stable end point and values from such groups are used for further palaeomagnetic determinations.

All thermal demagnetization was conducted in a shielded room (Patton 1967) in which the field gradient was less than 200 gammas. The field was further reduced by orthogonal sets of Helmholtz coils. From monitoring of the field before and after thermal cleaning, it was found that the field rarely exceeded 25 gammas during heating and cooling and usually was less than 10 gammas.

Most of the thermal demagnetization curves (Fig. 3) show that the specimens lose their magnetization through a range of temperatures. Only a few (Fig. 3(d)) show a narrow range of blocking temperatures. The Curie temperature indicated for the red beds is about 675 °C which corresponds to the known Curie temperature of haematite (Nagata 1961). The yellow beds rapidly lose their remanence at low temperatures, but often gain in intensity at 350 °C with a concordant change in colour from yellow to red. Since goethite has a Curie temperature between 97 °C (Hrynkiewicz, Kulgazuk & Tomala 1965) and 120 °C (Strangway *et al.* 1968) and inverts to haematite at 350 °C, it appears that most of the remanence in the yellow beds is carried by geothite.

Ore mineralogy of the Navajo sandstone

After the pilot specimens were examined, a representative number of sites from normal, reversed, and intermediate magnetic directions were chosen for petrographic analysis. The petrographic work is from a total of 40 thin sections and 21 heavy mineral mounts. The samples chosen for thin sections were examined on a standard petrographic microscope with a mechanical stage. A minimum of 500 mineral grains were identified and counted. The heavy minerals were divided into an opaque and non-opaque group. The heavy mineral mounts were made in the standard manner (Milner 1952).

The important features of the thin section work are first, that the percentages of total heavy minerals for the Navajo are higher than similar values reported for the same formation elsewhere (Harshbarger *et al.* 1957). The other important feature is that in the red haematite-stained, or red beds, the percentage of opaque heavies is generally higher than the percentage of non-opaque heavies.

In the red beds studied, the dominant iron oxide is haematite which occurs as discrete opaque grains, as minute particles imparting a red stain on other detrital

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grains, and as matrix material. In some cases it occurs as halos around weathered relics of pre-existing iron-minerals. Within the red-beds magnetite has not been identified by petrographic techniques, by thermal demagnetization experiments, or by X-ray analysis.

By volume the greater amount of haematite occurs as coatings on detrital grains and as colouring matter mixed with the matrix material. Staining material on detrital grains of quartz is nearly universal in the red beds. Where secondary quartz overgrowths are present haematite staining is found beneath the overgrowth. The haematite in the matrix material is found in the pore spaces between grains and at various types of grain contacts. Of the four types of grain contact, tangential, straight, concave, convex, and sutured (Sevier 1959), haematite may be found at any or all, but a larger percentage of tangential and straight grain contacts are likely to have haematite present.

Discussion of ore mineralogy

Dapples (1959) lists a typical diagenetic sequence for silica in the quartzose sandstones as:

(1) Rounding and solution;

(2) Precipitation of quartz overgrowths;

(3) Alteration of texture with greater burial;

(a) addition of carbonate cement, (b) solution of quartz at carbonate contacts, (c) sutured quartz grain contacts.

Many others (Folk 1951; Correns 1950; Waldschmidt 1941) suggest similar sequences. Since this sequence is present in these red beds, a time relationship for the introduction of the haematite may be deduced. The distribution of the haematite staining, although not indicative of primary or secondary origin, is sufficient for placing some time limits on its introduction. Since haematite is present beneath the secondary overgrowths and at some sutured contacts, some haematite must have been present in these rocks during early diagenesis. Therefore if only one generation of haematite is present, that haematite must have been introduced sometime before diagenesis became advanced, i.e. during the Jurassic, and this is probably true of the relatively impermeable red beds of the Kayenta and Summerville formations. However, the red beds of the Navajo Sandstone are relatively permeable, and in this formation a second generation of haematization can be found. This later haematite is found as staining over secondary overgrowths, as pore fillings, and as staining material over relatively clear calcite cement. This indicates that some haematite was added to this formation during later diagenesis or during relatively recent gelogic time. The volume of this later haematite appears to be greater than the volume of earlier haematite.

The yellow coloured sandstones are closely-related stratigraphically and genetically to the red beds. Indeed the yellow beds are found only associated with the red beds. The colour in these yellow beds may follow bedding and parting planes or it may cut across these features suggesting that the colour of these rocks is a secondary feature.

Since all the palaeomagnetic field directions were from structurally folded rocks, all directions reported in Table 1 are corrected to the present horizontal (*in situ*) and the palaeohorizontal. Site mean directions are shown in the following figures and formation means, or mean directions derived from a number of site means from within a formation, are reported in Table 1.

The site mean directions from the Kayenta Formation (Fig. 4) yield normal as well as reversed directions. The Kayenta directions all had low dispersion and high

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Table 1

Palaeomagnetic site mean directions, Fisher statistics, and virtual pole positions.

	Formation	Average declination	Average inclination	Virtual longitude	Pole latitude	N	R	K	α_{95}
1a.	Kayenta corrected to Jurassic Horizontal	165·5	−17·1	82·5E	61·2N	7	6·93	101·9	6·0
1b.	Kayenta corrected to present horizontal	173·1	−29·4	63·9E	53·5S	7	6·94	87·0	6·5
2a.	Navajo (all) 100 site means Jurassic horizontal	359·5	45·9	71·1E	79·98N	10	9·87	72·3	5·7
2b.	Navajo (all) present horizontal East Limb of Fold	4·0	55·1	5·2E	86·3N	10	9·95	195·1	5·7
3a.	Navajo (10) Jurassic horizontal	347·8	35·6	100·8	69·4	10	9·77	78·3	5·5
3b.	Navajo (10) Present horizontal	358·8	54·9	93·4	87·9	10	9·77	78·3	5·5
4a.	Navajo (10) Jurassic horizontal	13·1	49·4	5·4	77·1	10	9·81	196·9	3·5
4b.	Navajo (10) Present horizontal	4·6	53·6	15·3	85·1	10	9·93	210·9	3·3
5a.	Navajo (20) Jurassic horizontal both limbs of fold	359·0	43·2	70·4	77·8	20	19·44	34·0	5·7
5b.	Navajo (20) Present horizontal both limbs of fold	1·8	54·3	36·0	87·2	20	19·69	113·5	3·1
6a.	Navajo reversed Jurassic horizontal	173·0	−17·5	65·3E	46·45	25	19·4	4·2	15·9
6b.	Navajo reversed Present horizontal	170·2	−9·0	65·7E	43·25	25	18·9	3·9	16·8

stability after thermal cleaning at temperatures greater than 350 °C. Since dispersion is lessened by correcting to the palaeohorizontal, the Kayenta directions by the fold test were acquired pre-folding. Since the virtual pole position from the Kayenta of Gunlock is

82·5 East Longitude, 61·2 North Latitude,

which is close to the virtual pole position of

98 East Longitude, 59 North Latitude,

reported by Helsely & Steiner (1972) from the Kayenta near Moab, Utah, and since the Kayenta yields a virtual pole similar to other Triassic poles for North America, its magnetization at Gunlock was probably acquired during the Late Triassic.

In contrast, the conformably superjacent Navajo Sandstone directions (Fig. 5), after correcting for the bedding tilt, has a virtual pole position of

72 East Longitude, 80 North Latitude.

This is close to the pole position estimated by Runcorn (1956) from the uncleaned Kayenta and Carmel formations, and by Opdyke & Wensink (1966) from Lower Jurassic–Upper Triassic volcanics of the Connecticut Valley. In the computations, sites in which the scatter was high, yielding a radius of the circle of confidence greater

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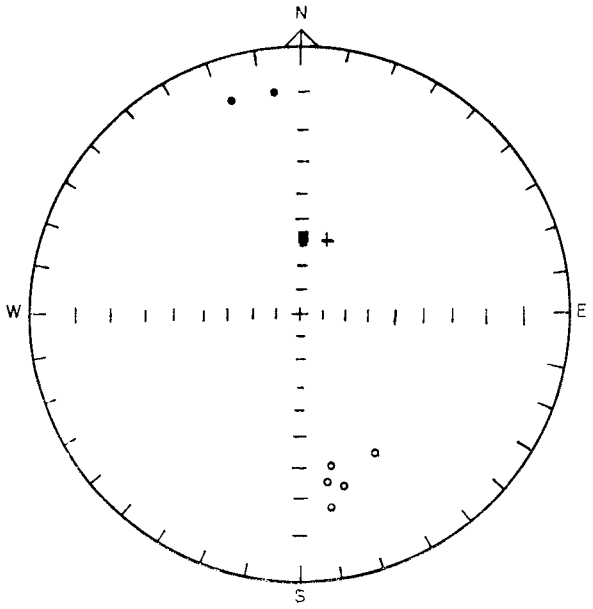


FIG. 4. Site mean directions Kayental formation corrected to Upper Triassic horizontal. Present-day field direction. Geocentric axial dipole field direction.

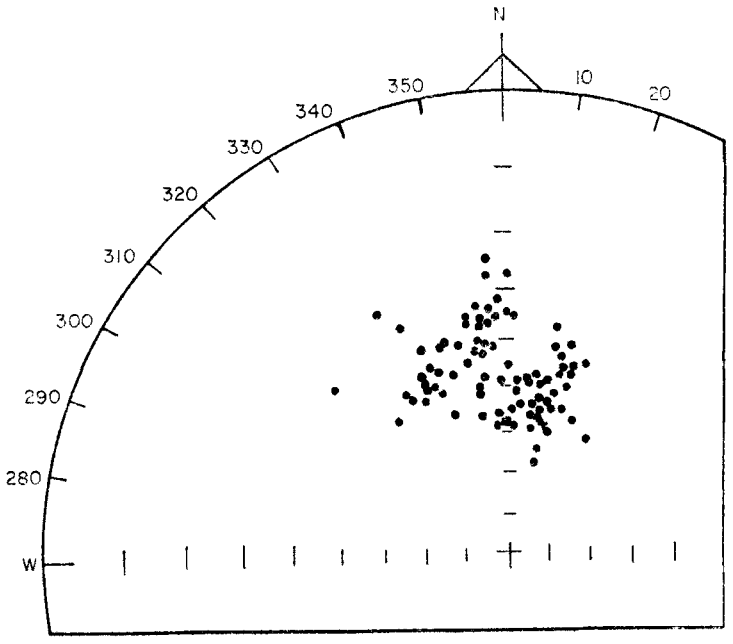


FIG. 5. Site mean directions Navajo sandstone, Gunlock, Utah corrected to Jurassic horizontal, only normal directions presented. Site mean direction. Present-day field direction. Axial dipole field direction.

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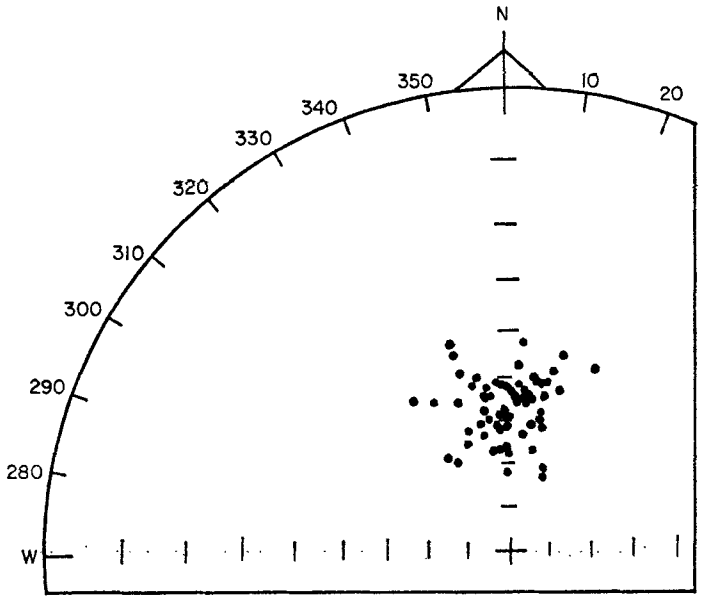


FIG. 6. Site mean directions Navajo sandstone, Gunlock, Utah corrected to present horizontal. Site mean direction. Present-day field direction. Axial dipole field direction.

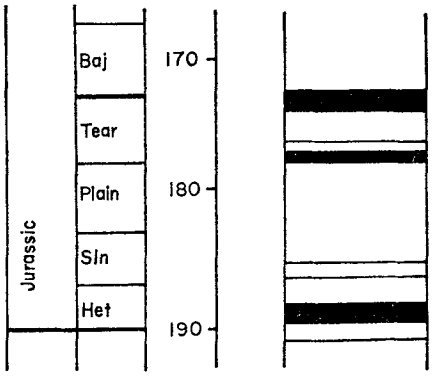


FIG. 7. Stratigraphic position of polarities encountered in Jurassic rocks from Utah. Normal polarity is unshaded: thickness of polarity intervals is approximately proportional to the stratigraphic thickness of the rocks found.

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than 20°, were excluded. The site means for a second group of Navajo samples from the San Rafael Swell yield a different virtual pole located at

355 East Longitude, 81·8 North Latitude.

The Navajo directions are nearly 20 degrees removed from the Kayenta directions. Such results are hardly compatible with the short time interval believed to exist between the deposition of the Kayenta and the Navajo. In terms of numbers it would require plate movements of nearly 37 cm/yr or more than twice the most rapid rate of spreading ever proposed.

One hundred Navajo site means from the Gunlock Section have therefore been replotted with the mean directions corrected to the present horizontal and the results are shown in Fig. 6. Although the results are from opposite limbs of a fold, they are from different stratigraphic horizons and therefore this is not strictly a fold test. In Table 1, the Navajo mean direction is derived by calculating a group mean from 10 site means and then calculating a formation mean from 10 group means. The fold test applied includes 20 closely-spaced stratigraphic sites, 10 sites from each limb of the fold. The results of 5(a) and (b) in Table 1 clearly show that by applying the fold test, one must conclude that the magnetization is post-folding.

The inescapable conclusion, therefore, seems to be that the stable magnetization is secondary and originated at a time when the magnetic field was not significantly different from the present, thus excluding the predominantly normally magnetized Cretaceous for the acquisition of magnetization. This raises the question that given a field of alternating polarity (Fig. 7), as found in this stratigraphic sequence, which is characteristic of the Cenozoic; why are the secondary magnetizations predominantly normal? It seems probable that the precipitation of iron was during the immediate past, i.e. during the Brunhes Normal. This implies a high mobility of the iron and the possibility of solution and redeposition of iron oxides on more than one occasion.

In an earlier paper (Johnson, Nairn & Peterson 1972) the attempt was made to establish a reversal chronology for the Jurassic, for when treated separately, the mean Navajo reversed site directions appear close to mean Kayenta reversed directions (Figs 8 and 9).

The mean direction of the reversed Navajo sites is quite close to the reversed Kayenta site means. In addition the reversed Navajo sites are far from the reversed present day pole position. Applying the fold test to these highly scattered reversed Navajo sites shows that the scatter is minimized after correcting to the palaeo-horizontal. Since the number of reversed Navajo samples is small, and the dispersion is large, detailed analysis is precluded. However, it seems possible that in the reversed samples the relics of an original Jurassic direction may not have been completely obliterated by the development of the stable secondary component. In addition other authors have reported reversals from equivalent or nearly equivalent stratigraphic levels from Jurassic rocks from other continents (Creer 1971; McElhinny & Burek 1971; Pechersky & Khramov 1973), which indicates that these reversals are Jurassic in origin.

Conclusions

From petrographic and magnetization studies it is concluded that the magnetization for both the Navajo and Kayenta is carried by the interstitial haematite or limonite. The magnetization is of chemical origin and thus accounts for its stability. The mean directions of magnetization of the few reversed scattered Navajo sites lie close to that derived for the Kayenta Formation. This suggests that these rocks may still possess an appreciable primary component of Jurassic origin. Logic suggests that some of the normally magnetized sites should possess an imperfect Jurassic com-

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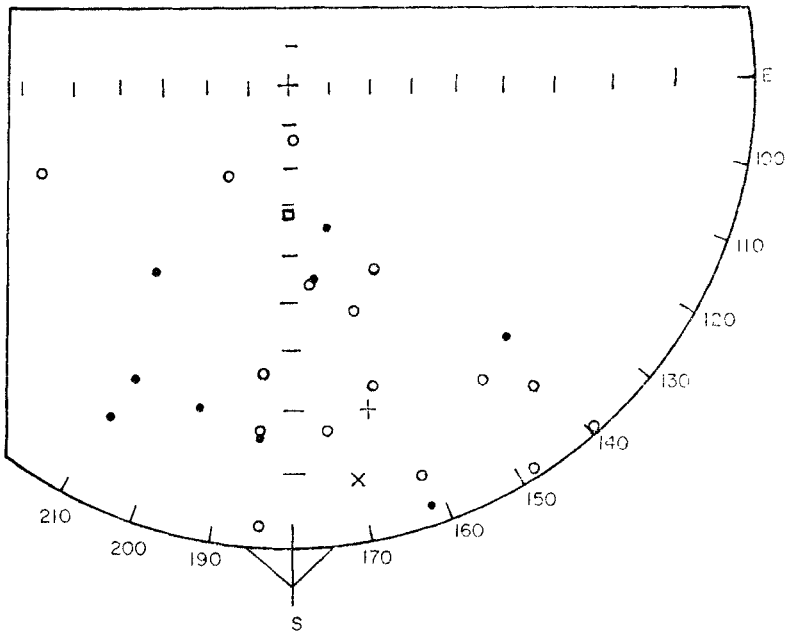


FIG. 8. Site mean directions reversed Navajo sandstone, Gunlock, Utah corrected to present horizontal. ●, Site mean directions (normal); ○, site mean directions (reversed); □, present-day reversed; +, mean Kayenta reversed; ×, mean Navajo reversed.

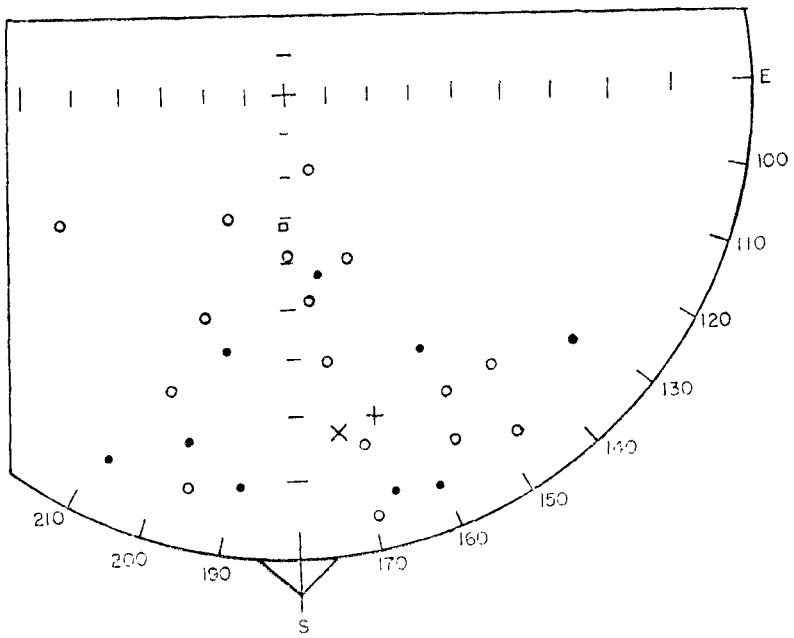


FIG. 9. Site mean directions reversed Navajo sandstone, Gunlock. Corrected to Jurrassic horizontal. Key as for Fig. 8.

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ponent, but unless an arbitrary decision is made to use a given value K, the dispersion, as a criterion, it is not realistic to make a distinction between them and the bulk of the normally magnetized sites.

The recent precipitation of iron during the immediate past, i.e. during the Brunhes Normal interval, implies considerable mobility of the iron. Two possible mechanisms seem to be likely means by which the mobility of the iron may be accomplished. A rapidly fluctuating water table, associated with the pluvial-interpluvials of the Pleistocene may account for the mobility. The iron would be dissolved during the high water levels of the pluvials and precipitated during the interpluvials. Of course this suggests also that the ground water must have been acidic and very low in dissolved oxygen. Since the retreat of the Wisconsin glaciation, the south-west has been in an interpluvial state (Anderson 1962). The second mechanism is that percolating ground water dissolves iron and carries it in solution. The iron is deposited wherever the ground water comes to the surface of the formation and evaporates. The second mechanism implies that the secondary magnetization is only superficial. Since deep drilling was not performed on these rocks, one cannot determine which explanation from palaeomagnetic or from geologic arguments is preferable.

These conclusions are speculative but follow from the conclusions based on the magnetic and petrographic study. Since the palaeomagnetic directions corrected to the Jurassic horizontal are broadly similar to previous work, erroneous conclusions could easily have been made if the sedimentologic study had not indicated that some of the haematite may be of secondary origin. The result from the reversed horizons suggest the field was little different from that of the Late Triassic, but it cannot be regarded as conclusive since there is no assurance that all secondary effects have been removed. This demonstrates the need for petrographic examination, a fold test, or both to accompany any magnetic study of porous sediments.

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